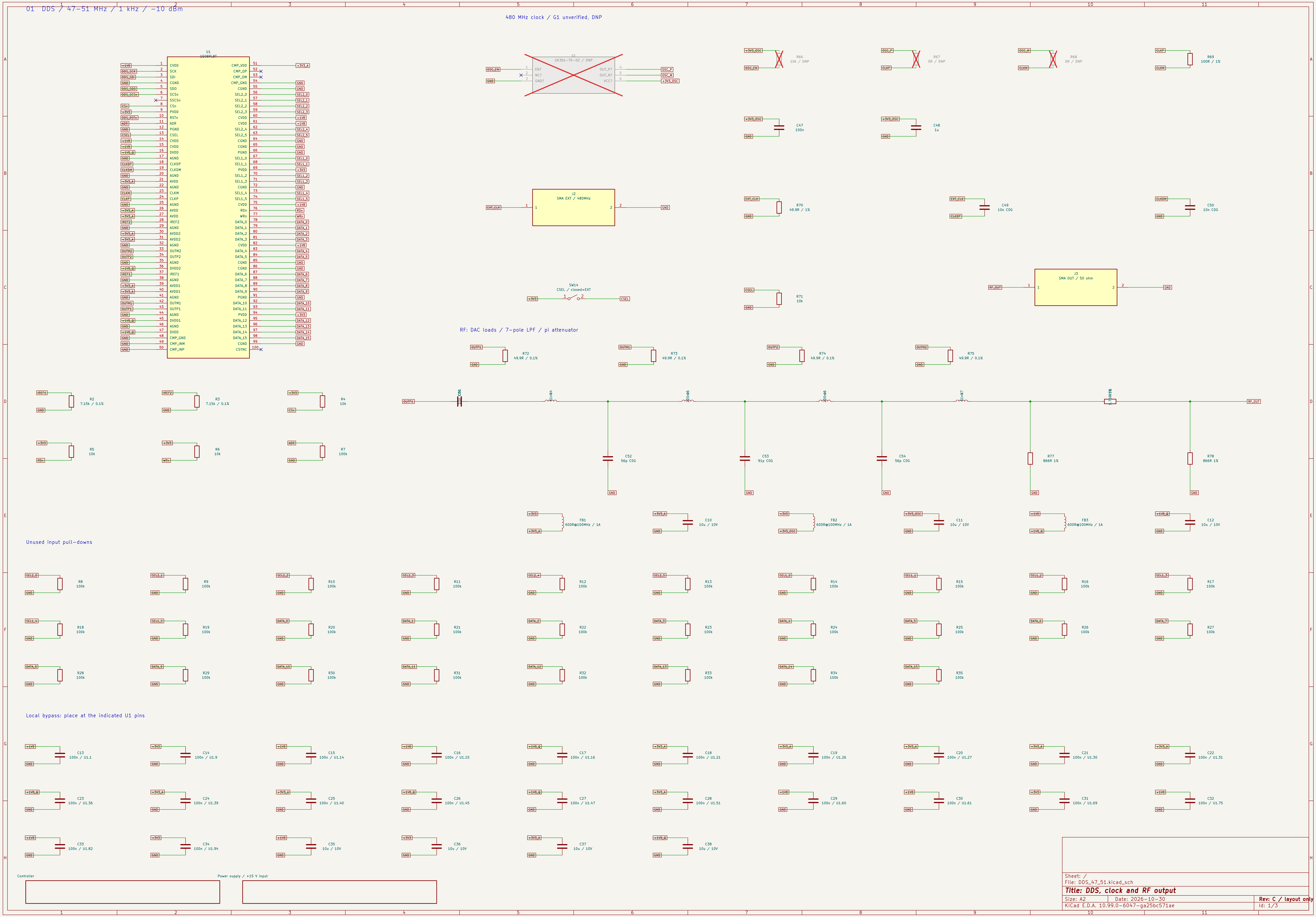

01 DDS / 47–51 MHz / 1 kHz / −10 dBm
U1
1508PL8T
1 CVDD +1V8
2 SCK DDS_SCK
3 SDI DDS_SDI
4 CGND GND
5 SDO DDS_SDO
6 SCSn DDS_SCSn
7 SSCSn
8 CSn CSn
9 PVDD +3V3
10 RSTn DDS_RSTn
11 ADR ADR
12 PGND GND
13 CSEL CSEL
14 CVDD +1V8
15 CVDD +1V8
16 DVDD +1V8_Q
17 AGND GND
18 CLKDP CLKDP
19 CLKDM CLKDM
20 AGND GND
21 AVDD +3V3_A
22 AGND GND
23 CLKM CLKM
24 CLKP CLKP
25 AGND GND
26 AVDD +3V3_A
27 AVDD +3V3_A
28 IREF2 IREF2
29 AGND GND
30 AVDD2 +3V3_A
31 AVDD2 +3V3_A
32 AGND GND
33 OUTM2 OUTM2
34 OUTP2 OUTP2
35 AGND GND
36 DVDD2 +1V8_Q
37 IREF1 IREF1
38 AGND GND
39 AVDD1 +3V3_A
40 AVDD1 +3V3_A
41 AGND GND
42 OUTM1 OUTM1
43 OUTP1 OUTP1
44 AGND GND
45 DVDD1 +1V8_Q
46 AGND GND
47 DVDD +1V8_Q
48 CMP_GND GND
49 CMP_INM GND
50 CMP_INP GND
51 CMP_VDD +3V3_A
52 CMP_OP
53 CMP_OM
54 CMP_GND GND
55 CGND GND
56 SEL2_0 SEL2_0
57 SEL2_1 SEL2_1
58 SEL2_2 SEL2_2
59 SEL2_3 SEL2_3
60 CVDD +1V8
61 CVDD +1V8
62 SEL2_4 SEL2_4
63 SEL2_5 SEL2_5
64 CGND GND
65 CGND GND
66 PGND GND
67 SEL1_0 SEL1_0
68 SEL1_1 SEL1_1
69 PVDD +3V3
70 SEL1_2 SEL1_2
71 SEL1_3 SEL1_3
72 CGND GND
73 SEL1_4 SEL1_4
74 SEL1_5 SEL1_5
75 CVDD +1V8
76 RDn RDn
77 WRn WRn
78 DATA_0 DATA_0
79 DATA_1 DATA_1
80 DATA_2 DATA_2
81 DATA_3 DATA_3
82 CVDD +1V8
83 DATA_4 DATA_4
84 DATA_5 DATA_5
85 CGND GND
86 CGND GND
87 DATA_6 DATA_6
88 DATA_7 DATA_7
89 DATA_8 DATA_8
90 DATA_9 DATA_9
91 PGND GND
92 DATA_10 DATA_10
93 DATA_11 DATA_11
94 PVDD +3V3
95 DATA_12 DATA_12
96 DATA_13 DATA_13
97 DATA_14 DATA_14
98 DATA_15 DATA_15
99 CGND GND
100 CSYNC
480 MHz clock / G1 unverified, DNP
G1
GK361-TK-02 / DNP
1 EN? OSC_EN
2 NC?
3 GND? GND
4 OUT_P? OSC_P
5 OUT_N? OSC_M
6 VCC? +3V3_OSC
R66 10k / DNP +3V3_OSC OSC_EN
R67 0R / DNP OSC_P CLKP
R68 0R / DNP OSC_M CLKM
R69 100R / 1% CLKP CLKM
C47 100n +3V3_OSC GND
C48 1u +3V3_OSC GND
J2
SMA EXT / 480MHz
1 EXT_CLK
2 GND
R70 49.9R / 1% EXT_CLK GND
C49 10n C0G EXT_CLK CLKDP
C50 10n C0G CLKDM GND
J3
SMA OUT / 50 ohm
1 RF_OUT
2 GND
SW14
CSEL / closed=EXT
+3V3 1 2 CSEL
R71 10k CSEL GND
RF: DAC loads / 7–pole LPF / pi attenuator
R72 49.9R / 0.1% OUTP1 GND
R73 49.9R / 0.1% OUTM1 GND
R74 49.9R / 0.1% OUTP2 GND
R75 49.9R / 0.1% OUTM2 GND
OUTP1
C52 56p C0G
C53 91p C0G
C54 56p C0G
R77 866R 1%
R78 866R 1%
RF_OUT
GND
R2 7.15k / 0.1% IREF1 GND
R3 7.15k / 0.1% IREF2 GND
R4 10k +3V3 CSn
R5 10k +3V3 RDn
R6 10k +3V3 WRn
R7 100k ADR GND
FB1 600R@100MHz / 1A +3V3 +3V3_A
C10 10u / 10V +3V3_A GND
FB2 600R@100MHz / 1A +3V3 +3V3_OSC
C11 10u / 10V +3V3_OSC GND
FB3 600R@100MHz / 1A +1V8 +1V8_Q
C12 10u / 10V +1V8_Q GND
Unused input pull–downs
R8 100k SEL2_0 GND
R9 100k SEL2_1 GND
R10 100k SEL2_2 GND
R11 100k SEL2_3 GND
R12 100k SEL2_4 GND
R13 100k SEL2_5 GND
R14 100k SEL1_0 GND
R15 100k SEL1_1 GND
R16 100k SEL1_2 GND
R17 100k SEL1_3 GND
R18 100k SEL1_4 GND
R19 100k SEL1_5 GND
R20 100k DATA_0 GND
R21 100k DATA_1 GND
R22 100k DATA_2 GND
R23 100k DATA_3 GND
R24 100k DATA_4 GND
R25 100k DATA_5 GND
R26 100k DATA_6 GND
R27 100k DATA_7 GND
R28 100k DATA_8 GND
R29 100k DATA_9 GND
R30 100k DATA_10 GND
R31 100k DATA_11 GND
R32 100k DATA_12 GND
R33 100k DATA_13 GND
R34 100k DATA_14 GND
R35 100k DATA_15 GND
Local bypass: place at the indicated U1 pins
C13 100n / U1.1 +1V8 GND
C14 100n / U1.9 +3V3 GND
C15 100n / U1.14 +1V8 GND
C16 100n / U1.15 +1V8 GND
C17 100n / U1.16 +1V8_Q GND
C18 100n / U1.21 +3V3_A GND
C19 100n / U1.26 +3V3_A GND
C20 100n / U1.27 +3V3_A GND
C21 100n / U1.30 +3V3_A GND
C22 100n / U1.31 +3V3_A GND
C23 100n / U1.36 +1V8_Q GND
C24 100n / U1.39 +3V3_A GND
C25 100n / U1.40 +3V3_A GND
C26 100n / U1.45 +1V8_Q GND
C27 100n / U1.47 +1V8_Q GND
C28 100n / U1.51 +3V3_A GND
C29 100n / U1.60 +1V8 GND
C30 100n / U1.61 +1V8 GND
C31 100n / U1.69 +3V3 GND
C32 100n / U1.75 +1V8 GND
C33 100n / U1.82 +1V8 GND
C34 100n / U1.94 +3V3 GND
C35 10u / 10V +1V8 GND
C36 10u / 10V +3V3 GND
C37 10u / 10V +3V3_A GND
C38 10u / 10V +1V8_Q GND
Controller
Power supply / +15 V input

Sheet: /
File: DDS_47_51.kicad_sch

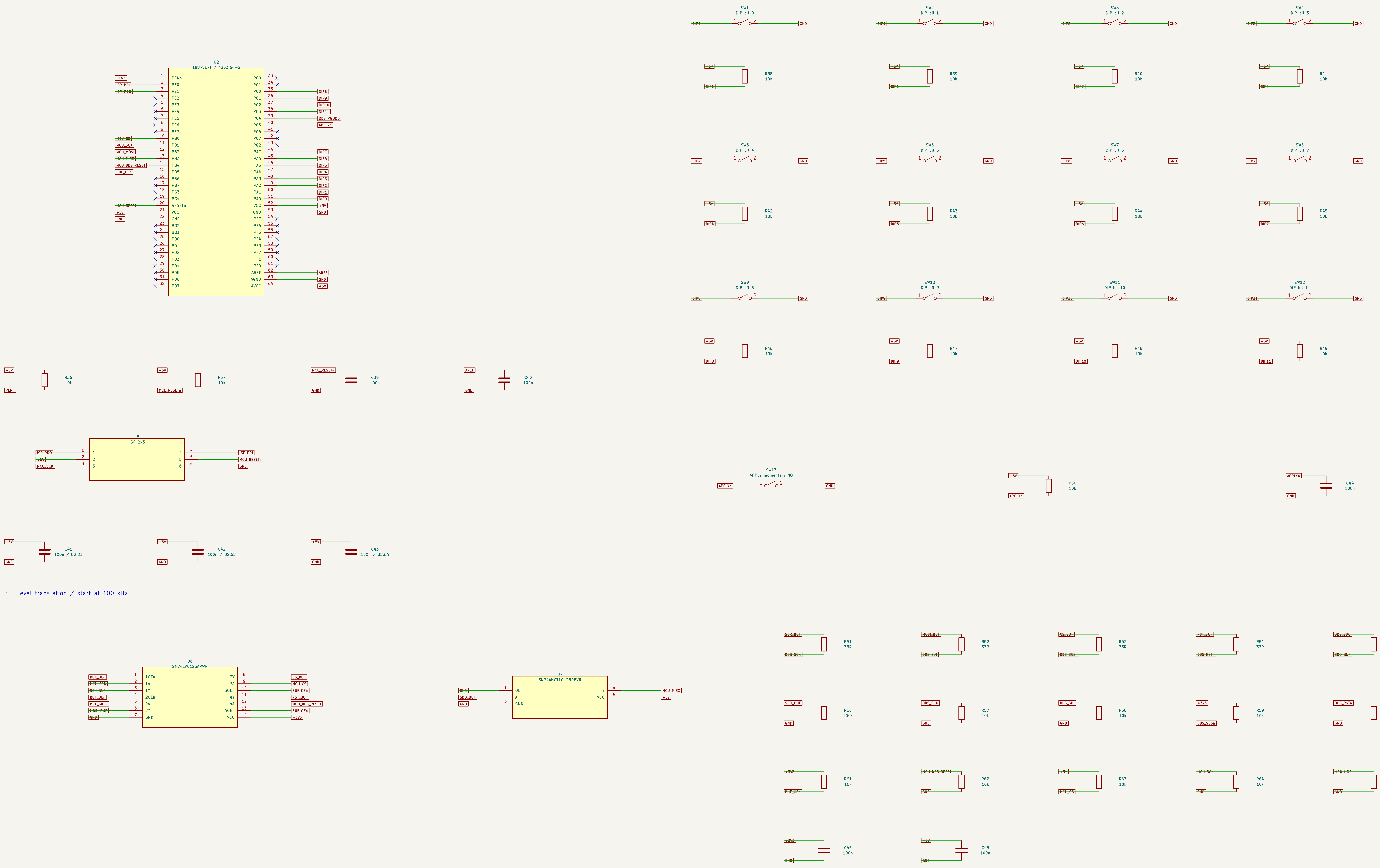
Title: DDS, clock and RF output

Size: A2 Date: 2026–10–30 **Rev: C / layout only / DRAFT**

KiCad E.D.A. 10.99.0–6047–ga25bc571ae Id: 1/3

02 CONTROLLER / 1887VE7T / +5 V

12-bit selector / ON = 1 / N = 0..4000


U2
1887VE7T / 4203.64-2
PENn
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PB0
PB1
PB2
PB3
PB4
PB5
PB6
PB7
PG3
PG4
RESETn
VCC
GND
BQ2
BQ1
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PG0
PG1
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
PG2
PA7
PA6
PA5
PA4
PA3
PA2
PA1
PA0
VCC
GND
PF7
PF6
PF5
PF4
PF3
PF2
PF1
PF0
AREF
AGND
AVCC
ISP_PDI
ISP_PDO
MCU_CS
MCU_SCK
MCU_MOSI
MCU_MISO
MCU_DDS_RESET
BUF_OEn
MCU_RESETn
+5V
GND
DIP0
DIP1
DIP2
DIP3
DIP4
DIP5
DIP6
DIP7
DIP8
DIP9
DIP10
DIP11
DDS_PGOOD
APPLYn
SW1
DIP bit 0
SW2
DIP bit 1
SW3
DIP bit 2
SW4
DIP bit 3
SW5
DIP bit 4
SW6
DIP bit 5
SW7
DIP bit 6
SW8
DIP bit 7
SW9
DIP bit 8
SW10
DIP bit 9
SW11
DIP bit 10
SW12
DIP bit 11
R38
10k
R39
10k
R40
10k
R41
10k
R42
10k
R43
10k
R44
10k
R45
10k
R46
10k
R47
10k
R48
10k
R49
10k
R36
10k
R37
10k
C39
100n
C40
100n
J5
ISP 2x3
C41
100n / U2.21
C42
100n / U2.52
C43
100n / U2.64
SW13
APPLY momentary NO
R50
10k
C44
100n
U6
SN74LVC125APWR
1OEn
1A
1Y
2OEn
2A
2Y
GND
3Y
3A
3OEn
4Y
4A
4OEn
VCC
SCK_BUF
MOSI_BUF
CS_BUF
RST_BUF
+3V3
U7
SN74AHCT1G125DBVR
OEn
A
GND
Y
VCC
SDO_BUF
R51
33R
R52
33R
R53
33R
R54
33R
R55
1k
DDS_SCK
DDS_SDI
DDS_SCSn
DDS_RSTn
DDS_SDO
R56
100k
R57
10k
R58
10k
R59
10k
R60
10k
R61
10k
R62
10k
R63
10k
R64
10k
R65
10k
C45
100n
C46
100n

SPI level translation / start at 100 kHz

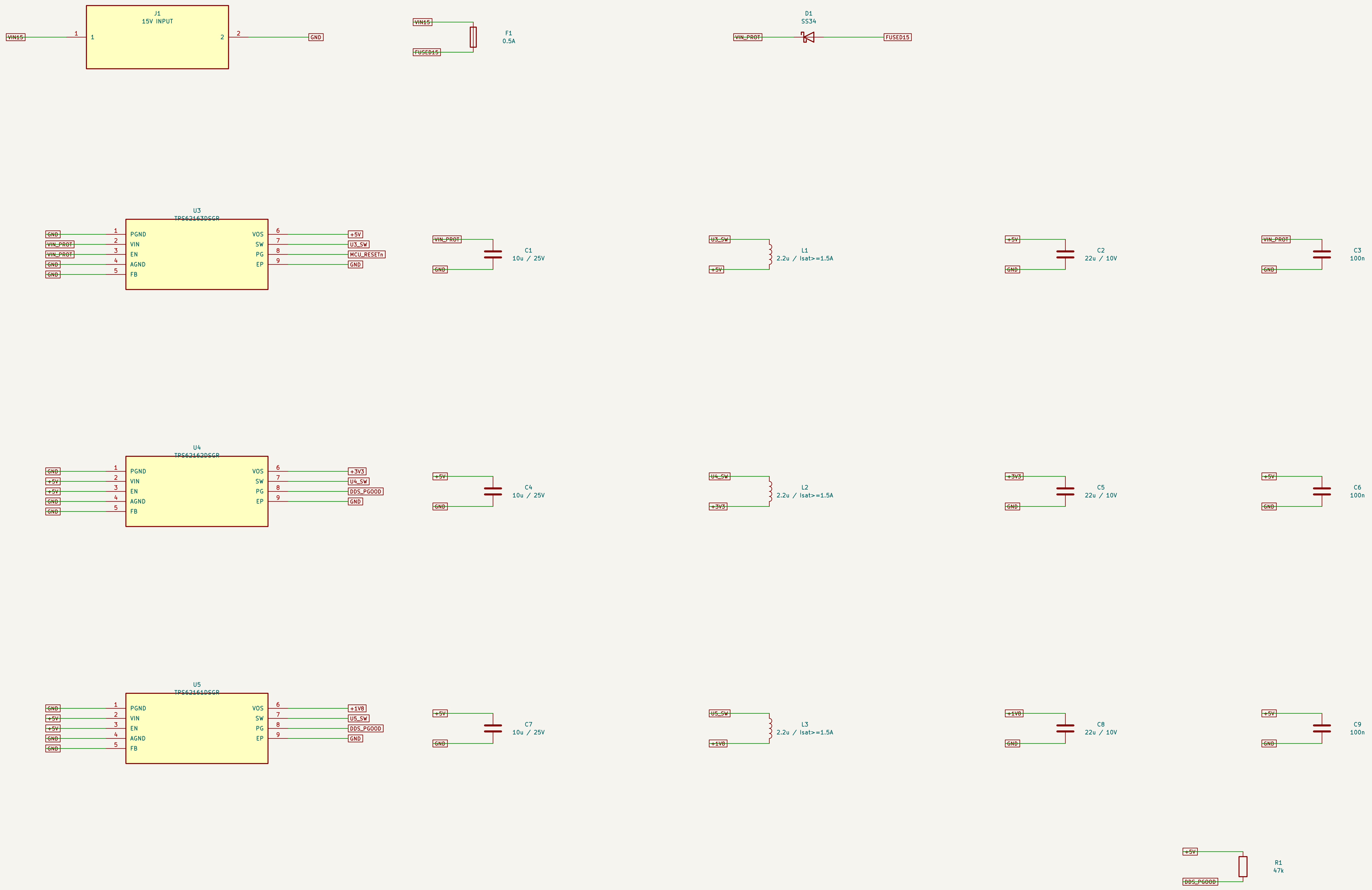
f = 47,000,000 + 1,000*N Hz. Reject N > 4000. Debounce APPLY.
Internal RC 8 MHz; ISP uses PE0/PE1. See design_notes.md for startup sequence.

Sheet: /Controller/
File: 02_controller.kicad_sch

Title: Controller, frequency selector and SPI

Size: A2
Date: 2026-10-30
Rev: C / layout only / DRAFT
KiCad E.D.A. 10.99.0-6047-ga25bc571ae
Id: 2/3

03 POWER / preliminary circuit


J1
15V INPUT
VIN15
1
2
GND
VIN15
FUSED15
F1
0.5A
D1
SS34
VIN_PROT
FUSED15
U3
TPS62163DSGR
GND
VIN_PROT
VIN_PROT
GND
GND
1 PGND
2 VIN
3 EN
4 AGND
5 FB
VOS 6
SW 7
PG 8
EP 9
+5V
U3_SW
MCU_RESETn
GND
VIN_PROT
GND
C1
10u / 25V
U3_SW
+5V
L1
2.2u / Isat>=1.5A
+5V
GND
C2
22u / 10V
VIN_PROT
GND
C3
100n
U4
TPS62162DSGR
GND
+5V
+5V
GND
GND
1 PGND
2 VIN
3 EN
4 AGND
5 FB
VOS 6
SW 7
PG 8
EP 9
+3V3
U4_SW
DDS_PGOOD
GND
+5V
GND
C4
10u / 25V
U4_SW
+3V3
L2
2.2u / Isat>=1.5A
+3V3
GND
C5
22u / 10V
+5V
GND
C6
100n
U5
TPS62161DSGR
GND
+5V
+5V
GND
GND
1 PGND
2 VIN
3 EN
4 AGND
5 FB
VOS 6
SW 7
PG 8
EP 9
+1V8
U5_SW
DDS_PGOOD
GND
+5V
GND
C7
10u / 25V
U5_SW
+1V8
L3
2.2u / Isat>=1.5A
+1V8
GND
C8
22u / 10V
+5V
GND
C9
100n
+5V
DDS_PGOOD
R1
47k

Input +15 V +/−5%; VIN <17 V. Effective capacitance at DC bias.
PG U4/U5 wired−AND. Verify oscillator current, thermal and noise performance.

Sheet: /Power supply / +15 V input/
File: 03_power.kicad_sch

Title: Power supply / +15 V input

Size: A3	Date: 2026−10−30	**Rev: C / layout only / DRAFT**
KiCad E.D.A. 10.99.0−6047−ga25bc571ae		Id: 3/3